

DISTANCE MEASURES IN N-DIMENSIONAL FEATURE SPACE

Dataset: Iris Dataset
 Number of samples : 150
 Number of features: 4

Features:
 1. sepal length (cm)
 2. sepal width (cm)
 3. petal length (cm)
 4. petal width (cm)

FIRST 10 SAMPLE POINTS

Sample	Sepal Length	Sepal Width	Petal Length	Petal Width
0	5.10	3.50	1.40	0.20
1	4.90	3.00	1.40	0.20
2	4.70	3.20	1.30	0.20
3	4.60	3.10	1.50	0.20
4	5.00	3.60	1.40	0.20
5	5.40	3.90	1.70	0.40
6	4.60	3.40	1.40	0.30
7	5.00	3.40	1.50	0.20
8	4.40	2.90	1.40	0.20
9	4.90	3.10	1.50	0.10

SELECT TWO SAMPLE POINTS

Enter index of first point (0-149): 0
 Enter index of second point (0-149): 1

Point P:
 [5.1 3.5 1.4 0.2]

Point Q:
 [4.9 3. 1.4 0.2]

Difference P - Q:
 [0.2 0.5 0. 0.]

DISTANCE MEASURES

1. Euclidean Distance
2. Manhattan Distance

3. Minkowski Distance
4. Chebyshev Distance
5. Squared Euclidean Distance
6. Cosine Distance
7. Calculate ALL distances
8. Exit

Enter your choice: 7

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DISTANCE RESULTS

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Euclidean Distance : 0.5385164807134502
Manhattan Distance : 0.6999999999999993
Chebyshev Distance : 0.5
Squared Euclidean Distance: 0.2899999999999997
Cosine Distance : 0.0014208364959781283
Minkowski Distance (r=3): 0.5104468722001463

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DISTANCE MEASURES

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1. Euclidean Distance
2. Manhattan Distance
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Enter your choice: 8

Program terminated.